PROJECT:NEURAL_NETWORKS

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# ==========================================
# HANDWRITTEN DIGIT RECOGNITION
# USING NEURAL NETWORK
# ==========================================

from sklearn.datasets import load_digits
from sklearn.model_selection import train_test_split
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score, classification_report

# ==========================================
# LOAD DATASET
# ==========================================

digits = load_digits()

X = digits.data
y = digits.target

# ==========================================
# TRAIN TEST SPLIT
# ==========================================

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

# ==========================================
# NEURAL NETWORK MODEL
# ==========================================

model = MLPClassifier(
    hidden_layer_sizes=(100, 50),
    activation='relu',
    max_iter=1000,
    random_state=42
)

model.fit(X_train, y_train)

# ==========================================
# PREDICTIONS
# ==========================================

predictions = model.predict(X_test)

# ==========================================
# EVALUATION
# ==========================================

accuracy = accuracy_score(y_test, predictions)

print("\nAccuracy:", accuracy)

print("\nClassification Report:")
print(classification_report(y_test, predictions))

# ==========================================
# TEST SAMPLE
# ==========================================

sample = X_test[0].reshape(1, -1)

result = model.predict(sample)

print("\nPredicted Digit:", result[0])
print("Actual Digit   :", y_test[0])
```

```
Accuracy: 0.9666666666666667

Classification Report:
              precision    recall  f1-score   support
```

```
           0       0.97      0.94      0.95        33
           1       0.96      0.93      0.95        28
           2       1.00      0.97      0.98        33
           3       0.97      0.97      0.97        34
           4       0.98      1.00      0.99        46
           5       0.96      0.96      0.96        47
           6       0.97      0.97      0.97        35
           7       1.00      0.97      0.99        34
           8       0.94      0.97      0.95        30
           9       0.93      0.97      0.95        40

    accuracy                           0.97       360
   macro avg       0.97      0.96      0.97       360
weighted avg       0.97      0.97      0.97       360


Predicted Digit: 6
Actual Digit   : 6
```